

LA ROCHELLE ILE DE RE AP, France

WMO: 073160

Lat: 46.1781N

Lon: 1.1931W

Elev: 23

StdP: 101.05

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.9	(c) -1.4	(d) -6.4	(e) 2.2	(f) -1.8	(g) -4.6	(h) 2.6	(i) -0.1	(j) 13.7	(k) 9.6	(l) 12.2	(m) 9.9	(n) 4.0	(o) 40	(p) 0.564

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.1	30.5	19.9	28.1	19.3	26.0	18.8	21.5	27.5	20.6	25.7	19.9	24.3	5.0	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.6	14.4	23.3	18.9	13.8	22.6	18.2	13.2	21.9	62.2	27.9	59.3	25.6	56.9	24.4	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.3	(b) 9.9	(c) 8.9	(e) -4.8	(f) 34.7	(g) 2.2	(h) 2.5	(i) -6.4	(j) 36.5	(k) -7.7	(l) 38.0	(m) -8.9	(n) 39.4	(o) -10.5	(p) 41.2		
(4) 11.3	(5) 9.9	(6) 8.9	(7) -4.8	(8) 34.7	(9) 2.2	(10) 2.5	(11) -6.4	(12) 36.5	(13) -7.7	(14) 38.0	(15) -8.9	(16) 39.4	(17) -10.5	(18) 41.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.2	6.6	6.9	9.4	11.9	14.8	18.3	20.3	19.6	17.8	14.8	10.2	7.3
	DBStd	5.82	3.60	3.65	2.98	2.70	2.73	3.23	2.81	2.62	2.97	3.17	3.41	3.75
	HDD10.0	406	111	94	46	10	1	0	0	0	0	4	39	99
	HDD18.3	2075	362	322	277	193	114	37	10	14	43	114	244	343
	CDD10.0	1569	7	6	27	67	150	249	320	298	234	152	45	14
	CDD18.3	198	0	0	0	1	4	37	72	53	27	4	0	0
	CDH23.3	1201	0	0	0	8	39	278	428	272	166	11	0	0
	CDH26.7	373	0	0	0	1	6	90	148	86	42	1	0	0
(14) Wind	WSAvg	4.7	4.6	5.0	5.0	4.6	4.8	4.6	4.7	4.6	4.3	4.3	4.8	4.8
(15) Precipitation	PrecAvg	735	74	56	50	61	51	36	44	47	56	83	96	80
	PrecMax	1016	166	118	162	240	88	85	130	109	210	173	259	148
	PrecMin	524	10	0	5	15	15	6	5	5	0	10	18	14
	PrecStd	131	40	32	36	45	21	18	23	32	47	42	50	38
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	16.2	21.3	24.6	27.2	33.3	33.9	32.6	30.9	24.7	18.8	14.7
		MCWB	12.2	11.2	13.7	15.9	18.6	21.6	21.6	20.4	19.7	17.8	15.2	12.7
	2%	DB	13.0	13.6	18.0	21.2	23.9	29.2	30.5	29.1	27.4	22.2	17.1	13.7
		MCWB	11.9	10.5	12.5	14.6	17.1	19.5	20.1	19.7	18.6	17.3	14.7	12.0
	5%	DB	12.1	12.3	15.6	18.9	21.5	26.3	27.9	26.2	25.0	20.6	16.1	13.1
		MCWB	11.1	10.2	11.2	13.5	16.2	18.7	19.5	19.2	17.6	16.6	14.1	11.8
	10%	DB	11.5	11.5	13.7	16.9	19.7	24.1	25.3	23.9	23.0	19.2	15.2	12.5
		MCWB	10.4	9.9	10.7	12.8	15.3	18.0	19.2	18.8	17.7	16.2	13.4	11.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.9	12.4	14.4	17.1	20.1	22.5	22.5	22.1	21.5	19.1	16.1	13.5
		MCDB	13.2	14.5	19.6	22.5	25.9	30.3	29.7	28.3	26.7	22.3	17.4	14.1
	2%	WB	12.1	11.5	12.9	15.4	18.0	20.9	21.4	21.1	20.1	18.1	15.2	12.7
		MCDB	12.6	12.7	16.7	19.8	22.1	26.9	27.7	26.2	24.3	20.5	16.5	13.3
	5%	WB	11.4	10.8	12.0	14.3	16.8	19.8	20.5	20.2	19.1	17.4	14.5	12.1
		MCDB	12.0	11.9	14.3	18.1	20.3	24.6	25.9	24.2	22.7	19.7	15.8	12.9
	10%	WB	10.6	10.1	11.3	13.2	15.9	18.7	19.8	19.5	18.3	16.7	13.5	11.2
		MCDB	11.3	11.3	13.1	16.2	18.9	22.8	24.6	22.9	21.7	18.8	15.0	12.3
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	6.2	7.2	8.2	7.9	8.8	8.1	8.4	9.2	7.4	6.2	5.5
		MCDBR	4.7	7.4	10.8	12.2	11.6	13.3	13.3	12.8	12.6	8.9	6.5	5.1
	5% WB	MCWBR	4.0	5.3	6.5	7.1	6.3	5.9	5.7	5.6	5.8	5.1	4.5	4.2
		MCWBR	4.1	5.9	8.2	10.4	9.6	11.3	10.6	9.7	9.3	7.1	5.5	4.5
(40) Clear-Sky Solar Irradiance	taub	taub	0.330	0.344	0.372	0.393	0.398	0.409	0.400	0.390	0.369	0.367	0.342	0.325
		taud	2.459	2.412	2.338	2.285	2.325	2.326	2.350	2.393	2.431	2.442	2.451	2.444
	Ebn at Noon	Ebn at Noon	766	827	850	863	870	859	862	857	846	788	741	725
		Edh at Noon	69	87	108	124	123	124	119	110	97	84	69	63
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.21	2.17	3.35	4.75	5.57	6.24	6.03	5.34	4.23	2.54	1.46	1.07
	RadStd	RadStd	0.12	0.22	0.37	0.45	0.43	0.48	0.29	0.38	0.28	0.21	0.16	0.13

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page